HAT-P-36b

R-0818

Benchmark run · workflow W8-benchmark-deep · record `deep-bench_HATP-36_250307` · created 2026-07-28T13:48:58+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2460741.8414 +/- 0.0042 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.11 +/- 0.013
Transit depth (Rp/Rs)^2	1.21 +/- 0.29 [%]
Orbital Inclination (inc)	83.4 +/- 2.37
Airmass coefficient 1 (a1)	1.423 +/- 0.018
Airmass coefficient 2 (a2)	-0.342 +/- 0.078
Scatter in the residuals of the lightcurve fit is	1.26 %
Best Comparison Star	#2 - [124, 183]
Optimal Aperture	3.46
Optimal Annulus	6.32
Transit Duration (day)	0.0771 +/- 0.0078

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.11 ± 0.013 vs 0.126 (deviation 0.88σ)
Duration fit vs published	0.0771 d vs 0.0925 d (deviation 1.41σ)
Mid-time O–C	9.9 min
Depth SNR	6.0
Residual scatter	1.26 %

Light curve

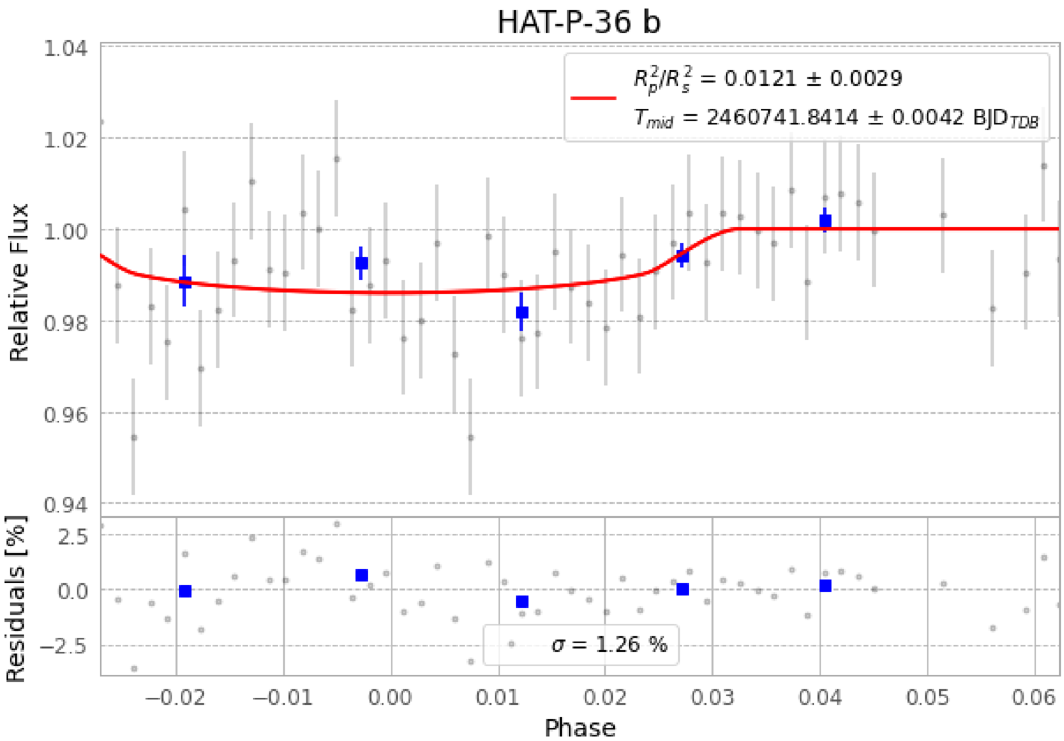

Figure 1 — FinalLightCurve_HAT-P-36 b_2025-03-07.png (SHA-256 8abd6e94fb5dd206…)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [396, 249], 3 comparison star(s). Exact configuration: `parameters.inits_json` in record.json (SHA-256 ee694837be6f51e6…).

`exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ead60b9`

Data provenance

54 FITS frames (35 MB), `HATP-36250307071215.FITS` → `HATP-36250307100315.FITS`. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-HATP-36-250307.log (a7c96bdbd7e0…), FinalLightCurve_HAT-P-36 b_2025-03-07.png (8abd6e94fb5d…), FinalLightCurve_HAT-P-36 b_2025-03-07.csv (321f5128e6c6…)

Compute resources

Wall time 120.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

